

Meeting Notes

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1 Meeting 1 [07/01/2026]

1.1 Agenda

- Validating the TER subject and obtaining an initial refocusing
- Knowing how to start the state of the art
- Choosing the tools to use
- Planning the next meeting

1.2 Topics Discussed

During the meeting, we validated the subject on partial preferences. The objective is really to explore this subfield of computational social choice theory by limiting ourselves to indivisible resources.

Afterwards, we listed some avenues for the state of the art. We will start by looking for articles on partial preferences focus on indivisible resources.

- Acceptability: Some researchers work on partial preferences under questions of acceptability by considering graphs (bipartite). For example, in the marriage problem, some agents refuse an allowance altogether. Thus, not all objects are allocable to a given agent. Acceptability therefore amounts to defining a list of preferences by excluding those on which the agent issues a right of veto.
- Additivity: Combining individual preferences into bundles of preferences adds information and complexity.
- Metric distortion problem: Ordinal preferences can be deduced from cardinal preferences and used as partial preferences. We then evaluate the extent to which the solution is degraded compared to what could have been proposed with the initial cardinal preferences.
- Cardinal preferences: Agents can express their preference in cardinal form, i.e. the intensity of the benefit they will derive from the allocation. Although this is suitable for evaluating equity criteria (maxmin, envy-free allocation), it is often difficult for agents to express such preferences.
- Truncated ballots: Derive from the vote theory.
- Uncertain preferences: The preferences are described by a probability distribution.

1.3 Tasks

For the next meeting on January 14, 2026, I will start the state of the art by listing partial/incomplete preference types.

2 Meeting 2 [14/01/2026]

2.1 Agenda

- Discussed on the partial preferences types found
- Clarifying some concepts
- Identifying key papers and resources
- Creating a GitHub repository for collaboration
- Planning the next meeting

2.2 Topics Discussed

During the meeting, we listed exhaustively the partial preferences. To prepare for the meeting, I had started reading the paper « Iterative voting with partial preferences » of Zoi Terzopoulou et al.

- Top k preferences: Agents only rank their top k choices, leaving the rest unranked.
- Truncated preferences: Similar to top k preferences, but the ranking is truncated at a specific point.
- Ordinal preferences: Agents express their preferences in a ranked order, without specifying the intensity of their preferences. It's a kind of distortion problem with respect to the cardinal preferences.
- Range of cardinal preferences: Agents provide a range of values for each option instead of a single value.
- Approval preferences: Agents indicate if a preference is approved or not, without ranking them. It's a binary indication and can be the result of boolean questions.
- Single peak preferences: Agents have a most preferred option, and their preference decreases as they move away from this peak (unimodal).
- Euclidean preferences: Agents' preferences are represented in a multi-dimensional space, where distance reflects preference intensity.
- Subset of possible orders: Agents provide a subset of possible rankings. The top k is a special case of this type.
- Combination of types: In practice, agents may express their preferences using a combination of the above types. For example, top k with structural aspects (single peak, euclidean).

Then, I learned how to search for articles on Google Scholar.

We also discussed researchers who have worked with partial preferences:

- Zoi Terzopoulou
- Sylvain Bouveret
- Jérôme Lang (director of LAMSADE): He studied the solutions that can occur and those that are valid (possible for a given problem).

Subsequently, we created a GitHub repository to collaborate on the TER.

Finally, we have also mentioned the value of partial information. In the problems considered, it is strongly assumed that agents clearly express their preferences. This work can be cognitively heavy for an agent who after a certain time will express himself at random. Then, sometimes the lack of information (for example on the ten candidates in an election) can prevent the expression of a complete ranking. How do you measure the value of a preference in a given context? This question is a matter of data elicitation and remains annexed to the study framework, although it justifies the importance of the study of partial preferences.

2.3 Tasks

For the next meeting on January 21, 2026, I will continue the state of the art by reading the articles.

3 Meeting 4 [03/02/2026]

3.1 Agenda

- Discussed following the reading of Jérôme Lang's survey and the chapter 10 of the Handbook of Computational Social Choice
- Updating the list of partial preferences
- Understanding the cross-domain aspect of "possible" and "necessary" properties.
- Identifying key papers and resources
- Planning the next meeting

3.2 Topics Discussed

To prepare for the meeting, I had started reading the paper the chapter 10 of the Handbook of Computational Social choice and the Jérôme Lang's survey "Collective Decision Making under Incomplete Knowledge: Possible and Necessary Solutions" (2020).

Updated list of partial preferences:

1. **Top k preferences:** Agents only rank their top k choices, leaving the rest unranked.
2. **Bottom-k preferences:** Agents only rank their bottom k alternatives, leaving the rest unranked.
3. **Truncated preferences:** Similar to top k preferences, but the ranking is truncated at a specific point. Truncated ballots.
4. **Ordinal preferences:** Agents express their preferences in a ranked order, without specifying the intensity of their preferences. It's a kind of distortion problem with respect to the cardinal preferences.
 - Ordinal preferences over the set of single goods, induced partial strict linear order over bundles.
5. **Range/Interval of cardinal preferences:** Agents provide a range of values for each option instead of a single value.
6. **Approval preferences:** Agents indicate if a preference is approved or not, without ranking them. It's a binary indication and can be the result of boolean questions. Agents approve a fixed number of alternatives or any number of alternatives.
7. **Veto preferences:** Agents disapprove a fixed number of alternatives or any number of alternatives. The k-veto is equivalent to (m-k)-approval preferences with m the number of alternatives.
8. **Full ranking preferences:** Agents provide a complete ranking of all alternatives.
9. **Next-best alternative queries:** Agent provide in sequential order her most preferred alternative among those not yet ranked.
10. **Top-k queries:** Agents approve k alternatives and give no order between them. Agents answer sequentially to the question "What are your most preferred alternatives?"
11. **Positional queries:** ask agent to provide the alternative she ranks at a given position t.
12. **Pairwise comparisons:** Agents indicate which of the both options considered is preferred.

13. **Single peak preferences:** Agents have a most preferred option, and their preference decreases as they move away from this peak (unimodal).
14. **Euclidean preferences:** Agents' preferences are represented in a multi-dimensional space, where distance reflects preference intensity.
15. **Subset of possible orders:** Agents provide a subset of possible rankings. The top k is a special case of this type.
16. **Choice queries:** Agents indicate which alternative in a subset S is most preferred. Kind of clustering approach.
17. **Set ranking:** Agents provide a ranking in a subset of alternatives.
18. **Split queries:** ask agent to partition a subset of alternatives into two or more subsets and order them such that all alternatives in a higher-ranked subset are preferred to all alternatives in a lower-ranked subset. k-approval is a special case where two blocks of size k and m-k are created.
19. **Combinaison of types:** In practice, agents may express their preferences using a combination of the above types. For example, top k with structural aspects (single peak, euclidean).
20. **Incomplete Set of preference profiles:** Only the preference profile of a subset of agents is known.
 - Application: The votes of nonmanipulators are fully specified, while no information is given for the manipulators.
21. **Truly incomplete preferences:** ???
22. **Probabilistic preferences:** Agents' preference profile follow a probabilistic distribution. Maximum Likelihood Estimator to estimate the distribution characteristics from a sample of preferences. An alternative can be more likely than others.
23. **Attribute preference:** Agents have different preference profile depending on the attribute considered (politics, economics, immigration).

We also discussed a few points:

- Considering only the possible winners and exclude the losers in certain situation without affecting the problem solving.
- Binary axis to represent political affiliation may be simplistic. Unimodal can be limited to describe the reality.
- Multiwinner: avoiding this topic for our state of the art. It is a field in extension.

3.3 Tasks

For the next meeting on February 9, 2026, I will continue the state of the art by searching for articles on partial preferences expressed as intervals. I will also prepare the discussion by reading some articles: Kenig et Kimelfeld (2019) on MLE; Pini et al. (2009); Xia and Conitzer (2011b). The objective is to determine whether it is interesting to investigate deeper in these topics and extract research questions. I would also like to write my notes neatly.